Exp:14
11/4/24

Packet sniffing and Raw sockets

Aim :-
To write a program using RAW sockets to implement packet sniffing.

Algorithm :-
create raw socket
1. Bind the socket to network interface
2. Receive network packets using the RAW socket
3. Extract the packet header information
4. Display the source IP, destination IP and protocol
5. Continue capturing packets & display the details.

Code :-

```
import socket
import struct
s = socket.socket (socket.AF_INET,
        socket.SOCK_RAW, socket.IPPROTO_IP)
print ("Packet sniffer started..")
while True:
    packet, addr = s.recvfrom(65565)
    ip_header = packet [0:20]
    iph = struct.unpack('!BBHHHBBH4s4s'
                        ip_header)
```

Input :-

network packets ~~transmitted~~ through the network

output :-

Packet sniffer started

Source IP : 192.168.1.10

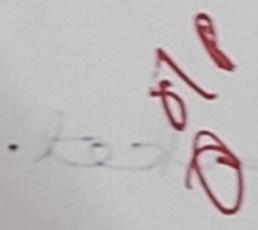
Destination IP : 142.250.183.14

Protocol : 6

Source IP : 192.168.1.10

Destination IP : 142.250.72.14

Protocol : 17



```
source_ip = socket.inet_ntoa(iph[8])
destination_ip = socket.inet_ntoa(iph[9])
protocol = iph[6]
print("Source IP :", source_ip)
print("Destination IP :", destination_ip)
print("Protocol :", protocol)
print("-" * 40)
```

Result :-

Thus, the program to implement packet sniffing using RAW sockets was executed successfully